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Owner **Erin Munsel:**
 Pediatric Clinical
 Pharmacist
 Specialist and
 Adjunct
 Area **Pharmacy
 Services**
 Applicability **UMHS Clinical**
 References **Guideline**

Pharmacy Services IV Medication Guidelines for Brandon Newborn Intensive Care Unit (NICU)

Drug Generic Name	Load/Bolus Dosing	Usual Dosing Range	Nursing Titration Instructions	Comments	Administration Concerns
ALPROSTADIL (PGE, PROSTIN, PROSTAGLANDIN) <i>*Used for PDA patency</i>	None	0.03-0.2 mcg/kg/min Hard max: 0.4 mcg/kg/min	DO NOT TITRATE	Common Side effects: apnea, fever	- Central line administration - If <1 mL/hr, D10W (or D5W) with heparin should be co-infused - If >1 mL/hr, no fluids needed for co-infusion
AMINOCAPROIC ACID [AMICAR®]	Loading: 50-100 mg/kg IV given over 15-60 minutes	25-30 mg/kg/hr Hard max: 100 mg/kg/hr	DO NOT TITRATE	Usually reserved for patients on ECMO with bleeding concerns	
CISATRICURIUM [NIMBEX®]	Loading: may give 0.1 mg/kg and then initiate continuous infusion	1-10 mcg/kg/min Hard max: 13 mcg/kg/min	DO NOT TITRATE	Dose adjustment ordered per patient activity/movement	Neuromuscular Blockade
DEXMEDETOMIDINE [PRECEDEX®]	Due to potential side effects, loading doses are not recommended	0.1-2 mcg/kg/hr Hard max: 2.5 mcg/kg/hr	DO NOT TITRATE	Common side effects: bradycardia and hypotension	
DOBUTAMINE	NONE	2.5-20 mcg/kg/min Hard max: 25 mcg/kg/min	Titrate by 2.5-5 mcg/kg/min every 5-10 minutes to maintain MAP within prescribed range.		Vesicant – if extravasation occurs, call Pharmacy (48208)
DOPAMINE	NONE	2.5 -20 mcg/kg/min Hard max: 25 mcg/kg/min	Titrate by 2.5-5 mcg/kg/min every 5-10 minutes to maintain MAP within prescribed range.		Vesicant – if extravasation occurs, call Pharmacy (48208)
EPINEPHRINE HYDROCHLORIDE	NONE	0.05-1 mcg/kg/min Hard max: 2 mcg/kg/min	Titrate by 0.01-0.02 mcg/kg/min every 5-10 minutes to maintain MAP within prescribed range.		Vesicant – if extravasation occurs, call Pharmacy (48208)

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EPOPROSTANOL [FLOLAN®]	NONE	2-20 ng/kg/ min Hard max: 125 ng/kg/ min Starting dose: 2-3 ng/kg/ min; increased based on patient response Typical dose increases by 1-2 ng/kg/ min 1 ng (nanogram) = 0.001 mcg	DO NOT TITRATE		- Flolan syringes must be changed: at least every 72 hours, when the drug syringe expires, or when the tubing needs changing per unit policy, which ever comes sooner - Protect from light - MUST BE FILTERED with 0.22-micron filter - Flolan is NOT COMPATIBLE with any other meds or IV fluids; run in dedicated line - DO NOT stop, pause, or interrupt infusion - Diluent: special Flolan diluent
ESMOLOL	Loading: 50-500 mcg/kg given over 1 minute	50-250 mcg/kg/ min Hard max: 800 mcg/kg/min	DO NOT TITRATE	MD order required for each dose change	No Diluent - straight drug
FENTANYL	Loading: 0.5-5 mcg/kg IV given slow IV push Administration rates loading/bolus doses: <5 mcg/kg = 3-5 minutes ≥5 mcg/kg = 5-10 minutes	1-5 mcg/kg/hr Hard max: 10 mcg/kg/hr	DO NOT TITRATE		- Pushing IV fentanyl too quickly may result in chest wall rigidity - The 50 mg/mL concentration has no diluent – straight drug
FUROSEMIDE	NONE	0.05-1 mg/kg/ hr Hard max: 1.5 mg/kg/hr	DO NOT TITRATE		The 10 mg/mL concentration has no diluent – this is straight drug
HEPARIN	Loading: NONE for neonates ECMO Loading: 100 units/ kg IV x 1	Initial: 10-20 units/kg/hr Hard max: 75 units/kg/hr ECMO: 10-50 units/kg/hr Hard max: 100 units/kg/ hr	DO NOT TITRATE	- Monitor aPTT or anti-Xa - Doses will be adjusted per levels - ECMO follows ACTs until stable, then transitions to anti-Xa monitoring	

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INSULIN REGULAR	NONE	Initiate at 0.05-0.1 units/kg/hr Hard max: 0.4 units/kg/hr	DO NOT TITRATE	Titrate by 0.01 units/kg/hr no more than every 2 hours to maintain ordered glucose range.	- MONITOR GLUCOSE LEVELS BRANDON HYPERGLYCEMIA GUIDELINES
LABETOLOL	Bolus: 0.2-1 mg/kg; may administer over 2 minutes	0.25-3 mg/kg/hr Hard max: 3 mg/kg/hr	DO NOT TITRATE		Concentrations greater than 2 mg/mL must go via central line
LIDOCAINE (VF/pulseless VT/shock)	1 mg/kg IV given over 2 minutes followed by continuous IV infusion; may administer second bolus if delay between initial bolus and start of infusion is >15 minutes	20-50 mcg/kg/min Hard max: 80 mcg/kg/min	DO NOT TITRATE		- Check lidocaine levels as requested by MD, space to Q24hr once therapeutic. - Therapeutic: 1.5-5 mcg/mL
MIDAZOLAM	Loading: 0.05-0.2 mg/kg IV given over 2 minutes	20-200 mcg/kg/hr Hard max: 300 mcg/kg/hr	DO NOT TITRATE		For refractory seizures, utilize EEG burst suppression orderable (*HIGH DOSE*)
MILRINONE	NONE	0.3-0.75 mcg/kg/min Hard max: 1 mcg/kg/min	DO NOT TITRATE	Side effects: hypotension	Renally eliminated, monitor for accumulation in renal dysfunction
MORPHINE SULFATE	Loading: 0.1-0.3 mg/kg IV given over 5 minutes	20-200 mcg/kg/hr Hard max: 300 mcg/kg/hr	DO NOT TITRATE		
NICARDIPINE	NONE	0.5-2 mcg/kg/min Hard max: 5 mcg/kg/min	DO NOT TITRATE		- Concentrations greater than 0.2 mg/mL must go via central line - Change peripheral infusion site every 12 hours to avoid irritation - If extravasation occurs, call Pharmacy (48208)
NOREPINEPHRINE BITARTRATE	NONE	0.05-1 mcg/kg/min Hard max: 2 mcg/kg/min	Titrate by 0.01-0.02 mcg/kg/min every 5-10 minutes to maintain MAP within prescribed range.		Vesicant – if extravasation occurs, call Pharmacy (48208)
OCTREOTIDE	GI bleed – may administer 1-2 mcg/kg IV load over 15-30 minutes; followed by continuous infusion Chylorthorax – None	GI Bleed: 1-10 mcg/kg/hr Chylorthorax: 1-6 mcg/kg/hr Hard max: 15 mcg/kg/hr	DO NOT TITRATE		
PANTOPRAZOLE	Loading: 1-2 mg/kg IV given over 5 minutes	0.1-0.2 mg/kg/hr Hard max: 0.5	DO NOT TITRATE		Should be used for GI bleeding only

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PENTOBARBITAL [NEMBUTAL®]	Loading: 1-5 mg/kg IV given over 20 minutes	mg/kg/hr 0.5-10 mg/kg/hr Hard max: 15 mg/kg/hr	DO NOT TITRATE		• Infusion, tubing, and stopcock needs to be changed Q24h • Does NOT need to be filtered
ROCURONIUM	NONE	5-10 mcg/kg/min Hard max: 12 mcg/kg/min	DO NOT TITRATE	Neuromuscular blockade Monitor with patient movement/activity	Vesicant – if extravasation occurs, call Pharmacy (48208)
SILDENAFIL	NONE	REFER TO SILDENAFIL GUIDELINES	DO NOT TITRATE		IV Sildenafil Guideline
VASOPRESSIN (CARDIAC SHOCK)	NONE	0.0001-0.002 unit/kg/min Hard max: 0.003 unit/kg/min	Titrate by 0.0001-0.0002 units/kg/min every 15-20 minutes to maintain MAP within prescribed range.	Pediatric dosing in shock states is in units/kg/min	Vesicant – if extravasation occurs, call Pharmacy (48208)

Approval Signatures

Step Description

Approver

Date

P&T	Karin Durant: Pharmacy Assistant Director and Adjunct Clinical I	1/19/2023
Newborn ICU Joint Practice	Michelle Nemshak: CLIN NURSE SPEC - (Peds-Perinatal) and Adjunct Cli	12/15/2022
PolicyStat Administrator Review	Olivia Curl: Admin Specialist Inter Health	11/21/2022
PolicyStat Administrator Review	Natalie Plata: Clinical Policy Manager	11/21/2022
Owner Review	Erin Munsel: Pediatric Clinical Pharmacist Specialist and Adjun	11/21/2022